Co-Op Presentation- 10-15min

- What we started with
- What we ended with
- How we pivoted data and edited prompts
- What we recommend not going forward with
- What this project could allow others to go forward with/how this can set up future co-ops
- Real world scenario link? - big picture w project
- Go through model and metrics- SHAWN
- Break down frameworks- Maddie- why they were chosen, their benefit, etc.
- Data and label discussion- Shweta
- Why was project necessary? Connect to ABIM mission
- Methodologies, Obstacles, work actually produced
- Results and future impact
- Gaps and Successes
- Biases and Limitations

Here is a suggested reorder for a typical project presentation:

1. Why was project necessary? Connect to ABIM mission
2. Where we started- how did we conclude this project was worth doing
3. Break down frameworks/my research - Maddie - why they were chosen, their benefit, etc.
4. How we pivoted data and edited prompts
5. Data and label discussion - Shweta
6. Methodologies, Obstacles, work actually produced
7. Go through model and metrics - SHAWN
8. What we ended with- Results
9. Future impact/Real world scenario link - big picture w project
10. Gaps and Successes
11. Biases and Limitations
12. What we recommend not going forward with
13. How can the future co-ops take this forward

Maddie's Role-

- Reviewed immersive job analysis on the general physician job. Focused specifically on the physician-patient intake assessment process.
    - Consists of: Developing treatment plans, making diagnoses, collecting/obtaining patient information, explaining procedures and discussing test results, etc.
    - Specifically chose an activity applicable to multiple scenarios/specialties
- Considered the soft and nuanced medical skills present in this activity
- Traced the skills and behaviors present in activity to competencies- patient care and interpersonal communication
- Found relevant frameworks - NURSE and Calgary - valid, reliable, and capable of being recognized by an LM
- Did immersive analysis on Frameworks
- Created checklists to determine if steps in frameworks are present in conversation
    - Absence vs. presence
- Helped out with determining and refining AI prompts to determine that the AI was actually capturing the intent of each framework
- Assisted with manual grading and developing the golden standard dataset
    - High quality vs. low quality conversations
- Focused on Inter-Rater Reliability Analysis- calculated consistency between human graders and the AI

Maddie
Slide 1-

- Currently, the only way we have to measure physician capability is through a test that only reaches and measures the competency of medical knowledge.
- Want to come up with ways to measure other competencies that are both lower in effort and efficient.
- However, human expert grading is expensive and time consuming, causing most doctors or training medical students to only be given feedback in limited amounts during entire training.
- Want to certify skills and attitudes, but have objectively been the hardest to measure.
- This project provides a potential solution to validate these “soft skills” by aligning with ABIM’s goal of ensuring diplomates are ensuring excellence in all facets of care, not just medical knowledge.

- Currently, our most robust metrics prioritize medical knowledge, yet we know that clinical excellence is as much about how a physician interacts as it is about what they know.
- Expert human grading of communication is the "gold standard," but its high cost and time intensity mean most trainees receive limited sporadic feedback across their entire career.
- While "skills and attitudes" are core to the ABIM mission, they have historically been the most elusive to measure objectively and at scale in the industry.
- We are validating an AI-driven model to turn nuanced relational skills into objective, actionable data, ensuring our diplomates excel in every facet of patient care to improve clinical outcomes.

MENTION-

- Physicians are busier than ever and one of the most important factors of their jobs, communication, is the hardest to measure and improve to current date.
    - Lack of time for trained and certified individuals to sit in and grade how trainees interact
    - Lack of a "bird’s-eye view" to determine standard of care
    - Lack of a standardized tool to make feedback objective. What one may consider a necessary demonstration of empathy, another might deem as unnecessary.
- We want to look for a way to address these issues while giving physicians the same data-driven insights into their communication that can improve clinical outcomes.
- Using Language Models (LMs) to act as a “second set of eyes” on patient interactions.

- Can we teach a machine to see patterns as accurately as a human?
- How did we decide on this project specifically?
- There is a massive verification gap in medicine. Certify doctors via exam but assume they are capable of proper human interaction.
- Limited ways to measure communication- only through patient satisfaction surveys -rare and few
- Current way to get certified is via summative feedback, want to move to formative
- Wanted to do something that measures a different competency outside of medical knowledge

Slide 1-

- Currently, our most robust metrics prioritize medical knowledge, yet we know that clinical excellence is as much about how a physician interacts as it is about what they know.
- Expert human grading of communication is the "gold standard," but its high cost and time intensity mean most trainees receive limited sporadic feedback, preventing the continuous mastery required for high-quality patient care.
- While "skills and attitudes" are core to the ABIM mission, they have historically been the most elusive to measure objectively and at scale, leaving a critical gap in professional accountability and standardization.
- We are validating an AI model that translates nuanced interaction into measurable data via standardized frameworks; this ensures our diplomates have the reliable tools needed to master the relational skills proven to improve clinical outcomes.

Here is a reordered outline, grouping related ideas and presenting them in a more logical flow from problem to solution:

Problem/Context

- Physicians are busier than ever and one of the most important factors of their jobs, communication, is the hardest to measure and improve to current date.
- Lack of time for trained and certified individuals to sit in and grade how trainees interact
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excellence is as much about how a physician interacts as it is about what they know.
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Motivation/Project Goal

- We want to look for a way to address these issues while giving physicians the same data-driven insights into their communication that can improve clinical outcomes.
- There is a massive verification gap in medicine. Certify doctors via exam but assume they are capable of proper human interaction.
- Limited ways to measure communication- only through patient satisfaction surveys -rare and few
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Proposed Solution

- Using Language Models (LMs) to act as a “second set of eyes” on patient interactions.
- Can we teach a machine to see patterns as accurately as a human?
- We are validating an AI model that translates nuanced interaction into measurable data via standardized frameworks; this ensures our diplomates have the reliable tools needed to master the relational skills proven to improve clinical outcomes.

Script-

"Now that we’ve established the LLM as our 'second set of eyes,' the next question was: Where do we point them? We chose to focus on the core physician encounter because it is the universal heartbeat of medicine, regardless of specialty. By mapping this encounter to ACGME milestones, we move from a simple experiment to a viable assessment tool."

SLIDE 1 final-

- Physicians are busier than ever, leaving little room to measure or improve how they communicate with patients.
    - Certified staff lack the time needed for direct observation and grading of trainees.
    - We lack the high level data needed to ensure a system-wide standard of care.
    - Without a standardized tool, one person’s “empathy” is another’s “inefficiency.”

SLIDE 2 final-

- Current assessments mainly measure medical knowledge, but true clinical excellence also depends on how physicians communicate with patients.
- Expert evaluation of communication is ideal, but it is too slow and costly to provide trainees with regular feedback.
- Professional skills and attitudes are highly valued, yet they remain the most difficult to measure consistently and fairly.
- As a result, there is still a gap in how we evaluate these important competencies.
- There is no scalable way to evaluate communication across large volumes of clinical conversations.

SLIDE 3 final-

- Approach: Utilize Language Models (LMs) as an automated "second set of eyes" to provide continuous, objective observation of patient interactions.
- The Research Question: Can AI-driven pattern recognition achieve the same level of nuance and accuracy as human expert evaluation?
- The Objective: Move from subjective interpretation to objective data by providing a consistent, standardized measure for clinical communication.
- The Ultimate Goal: Empower diplomates with evidence-based tools to refine the relational skills proven to improve care.

SLIDE 4 final-

- Main focus on the physician encounter as the primary unit of research.
- Why?
    - Ensures the tool is relevant to diverse medical specialties represented by the ABIM.
    - Creates a scalable model that works in any clinical setting or patient demographic.
    - Leverages the diverse and unique skills of the three Co-Ops to ensure a holistic view of the interaction.

SLIDE 5 final-

Points for myself

- Focused on the nuanced clinical skills that define a great physician.
- Identified 12 specific skills and organized them into five core categories: Emotional Core, Relational Mechanics, Information Gathering, Cognitive Integration, and Collaborative Planning.
- Mapped directly to ACGME Milestones and ABMS Competencies to ensure professional relevance.
- This transforms our experiment into a practical, standardized assessment tool.

SLIDE 6 final-

- Selected the **Calgary-Cambridge**, **NURSE**, and **SHARE** frameworks as the optimal evidence-based models to turn general communication into specific, repeatable bedside skills.
- Validated, reliable structures suitable for Language Model integration
- Addresses the three essential pillars of modern practice: structure, empathy, and collaboration.

SLIDE 7 final-

- A five-stage framework used to guide physicians through the chronological flow of a medical encounter.
- Integrates the fundamental skills of Building Relationships and Providing Structure throughout the entire patient encounter.
- Prioritizes a patient-centered approach to ensure the patient's perspective and concerns are fully uncovered.
- Research-validated steps shown to improve patient safety and health outcomes.

SLIDE 8 final-

- Empathy-based approach designed to acknowledge and validate patient emotions through a clinical lens.
- A practical tool used to lower patient stress levels and foster a more productive clinical environment.

- Requires the physician to pause and identify specific verbal and non-verbal cues from the patient.
- Serves as the primary mechanism for establishing trust and strengthening the patient-physician bond.

SLIDE 9 final-

- Positions the medical consultation as a collaborative partnership.
- Ensures a clear and thorough explanation of all available clinical pathways, including associated risks and benefits.
- Identifies the patient's unique priorities and personal values.
- Concludes only when both the physician and patient reach an agreement on a care plan that inspires mutual confidence.

- Reviewed preexisting immersive job analyses for both general physicians and those specifically belonging to internal medicine
- Group decision made to specifically focus on the physician encounter (the core consultation)
- Why?
    - It is a core encounter that has universal applicability across medical specialities
    - We wanted something that can be applicable to several areas and specialties.
    - Also wanted to do something that utilizes the three co-ops skills.
    -

Slide 2-

- Within this activity, we wanted to consider the soft and nuanced medical skills present
    - 5 categories of skills: The emotional core, relational mechanics, information gathering, cognitive integration, and the collaborative plan
    - Started with 16 skills, but ended with 12 after considering which ones are capable of being identified by an AI
- Use these skills and trace them back to ACGME Milestones and Competencies
- Why?
    - Turns this experiment into a potential assessment tool

- Main competencies present: Interpersonal Communication Skills, Patient Care, and Professionalism

Slide 3-

- Shows slides of mapping to milestones and competencies.

Slide 4-

- Used the gather information regarding competency mapping and milestone mapping to find relevant frameworks
- Reviewed preceding analyses on Calgary Cambridge, NURSE, and SHARE Frameworks to bridge gap between skill search
    - Use of these specific frameworks for two major reasons:
    - 1. Valid, reliable, and capable of being recognized by an LM
    - 2. Together, they cover the structural, emotional, and collaborative pillars of modern medicine

Slide 5-

- What is the Calgary Cambridge Model?
    - A five-stage framework used to guide physicians through the chronological flow of a medical encounter.
    - includes 2 continuous skills—Building Relationship and Providing Structure
    - Patient-centered focus- want to uncover patient agenda
    - Strongly supported by evidence- research shows that these specific communication steps improve patient health outcomes and safety

- What is the N.U.R.S.E. Framework?
    - Empathy based approach meant to acknowledge and validate a patients emotions
    - A descelation tool used to lower the patient's stress level
    - Requires the physician to stop and listen for cues
    - Primary tool used for building a relationship with the patient

- What is the S.H.A.R.E. Framework?
    - Allows the doctor to pose the consultation as a collaborative partnership

    - Requires the clinician to clearly explain all options and the risks and benefits associated with them
    - A core goal of finding out what matters most to the patient
    - Ends only when both the physician and patient reach a mutual agreement on a plan they both feel confident in.

- Only ended up following through with Calgary and NURSE
- Why?
    - Lack of time
- Why these specific two frameworks?
    - Strongly integrated together. NURSE has placeholders in Calgary
    - Calgary accounts for the largest amount of skills
    - Still accounts for all 12 skills
    - Avoid an overlap that would take place if kept share instead of NURSE
- Finally, created a checklist for the two frameworks
- Why?
- Creates a bridge between human nuance and machine logic
- Solves the fact that some concepts like “empathy” are too abstract to be picked up by a LM
- Creates specific verbal markers for the LM to pick up
- Creates pattern matching for the LM
- Minimized hallucination and bias- forces model to look for evidence

List of soft and nuanced medical skills in sources-

1. Empathy, Compassion, and Emotional regulation
2. Behavioral adaptability and patient centeredness
3. Communication
4. Relationship and Trust building
5. Contextual navigation
6. Complex clinical reasoning
7. Active listening
8. Shared decision making
9. Bridging relational gaps
10. Bedside Manner
11. Collaborative phrasing
12. Nonverbal facilitation

- Patient-Centered Interviewing and active listening
    - Nonverbal facilitation
    - Eliciting context
- Strategic agenda setting
    - Persistent open ended questioning
- Navigating and responding to emotion
    - Indirect emotional inquiry
- Shared decision making

SUGGESTED LIST OF SKILLS-

The 16 Nuanced Medical Skills

1. **Emotional Regulation:** Managing one's own internal state and biases to remain present and professional during stressful encounters.
2. **Empathy and Compassion (navigating and responding to emotion):** The foundational ability to perceive, share, and act upon the patient's emotional state.
3. **Behavioral Adaptability**: Dynamically adjusting your communication style, tone, and energy to match the patient's personality or cultural needs.
4. **Bedside Manner:** The collection of professional etiquette and physical presence (posture, eye contact) that establishes immediate rapport.
5. **Strategic Agenda Setting:** Negotiating the goals of the visit early on to balance the patient's concerns with medical priorities.
6. **Patient-Centered Interviewing:** Shifting the focus from a "checklist" history to a narrative-driven dialogue where the patient's story leads.
7. **Active Listening:** Engaging in a "loop" of listening, processing, and reflecting back to ensure the patient feels fully heard.
8. **Nonverbal Facilitation:** Using body language, silence, and "minimal encouragers" (nodding, leaning in) to prompt the patient to continue.
9. **Persistent Open-Ended Questioning:** Resisting the urge to ask "Yes/No" questions, instead using broad prompts to uncover deeper clinical data.
10. **Eliciting and Navigating Context:** Specifically seeking out the "real life" factors (finances, family, home) that influence a patient's health.
11. **Indirect Emotional Inquiry:** Using subtle "door-openers" to address emotions that the patient may not be stating directly.
12. **Bridging Relational Gaps**: Actively working to overcome barriers caused by differences in background, education, or previous medical trauma.

13. **Complex Clinical Reasoning:** The high-level synthesis of medical knowledge with the specific nuances and context of the individual patient.
14. **Collaborative Phrasing:** Using "we" language and inclusive terminology to transition from "doctor's orders" to a partnership.
15. **Shared Decision Making:** The formal process of weighing medical options against patient values to reach a mutually agreed-upon plan.
16. **Relationship and Trust Building:** The deliberate cultivation of a safe, reliable, and non-judgmental space for the patient.

Nuanced Medical Skill	ACGME Milestone (Sub-Competency)	ABIM Core Competency
1. Emotional Regulation	**PROF1:** Professional Behavior & Ethical Principles	**Professionalism**
2. Empathy & Compassion	**ICS1:** Patient- and Family-Centered Communication	**Interpersonal & Communication Skills**
3. Relationship & Trust Building	**PROF1 / ICS1:** Creating therapeutic alliances	**Professionalism**
4. Behavioral Adaptability	**PROF2:** Accountability & Adaptability	**Professionalism**
5. Bedside Manner	**ICS1:** Patient-centered communication	**Interpersonal & Communication Skills**

6. Strategic Agenda Setting	**ICS1:** Negotiating shared goals for the visit	**Interpersonal & Communication Skills**
7. Patient-Centered Interviewing	**PC1:** History; **ICS1:** Patient-centered interview	**Patient Care / Interpersonal Skills**
8. Active Listening	**PC1:** History (gathering accurate data)	**Patient Care**
9. Nonverbal Facilitation	**ICS1:** Verbal and nonverbal communication	**Interpersonal & Communication Skills**
10. Persistent Open-Ended Questioning	**PC1:** History (comprehensive and nuanced)	**Patient Care**
11. Eliciting & Navigating Context	**SBP3:** System Navigation for Patient-Centered Care	**Systems-Based Practice**
12. Indirect Emotional Inquiry	**ICS1:** Advanced responding to emotion	**Interpersonal & Communication Skills**
13. Bridging Relational Gaps	**PROF1:** Cultural humility and sensitivity	**Professionalism**

14. Complex Clinical Reasoning	**PC3:** Clinical Reasoning & Differential Diagnosis	**Medical Knowledge / Patient Care**
15. Collaborative Phrasing	**ICS1:** Use of inclusive/collaborative language	**Interpersonal & Communication Skills**
16. Shared Decision Making	**ICS1 / SBP3:** Collaborative management	**Patient Care**

Soft Skill Category	**ACGME Milestone (Sub-Competency)**	**ABIM Core Competency**
The Emotional Core 1. Emotional Regulation 2. Empathy & Compassion	**PROF1:** Professional Behavior and Ethical Principles; **ICS1:** Patient- and Family-Centered Communication	**Professionalism:** Maintaining an ethical and sensitive practice.

Relational Mechanics 3. Relationship & Trust Building 4. Bridging Relational Gaps 5. Bedside Manner 6. Behavioral Adaptability	**ICS1:** Patient- and Family-Centered Communication; **PROF1:** Cultural Humility and Sensitivity	**Interpersonal & Communication Skills:** Creating a therapeutic relationship.

Information Gathering 7. Active Listening 8. Persistent Open-Ended Questioning 9. Nonverbal Facilitation 10. Patient-Centered Interviewing	**PC1:** History; **ICS1:** Verbal and Non-verbal Communication Skills	**Patient Care:** Gathering accurate, context-rich data.
Cognitive Integration	**PC3:** Clinical Reasoning & Differential Diagnosis; **SBP3:** System Navigation	**Patient Care:** Synthesizing clinical and contextual data.

11. Complex Clinical Reasoning 12. Eliciting & Navigating Context		
The Collaborative Plan 13. Shared Decision Making 14. Collaborative Phrasing 15. Strategic Agenda Setting	**ICS1:** Negotiating Shared Goals; **SBP3:** Patient-Centered System Navigation	**Systems-Based Practice:** Coordinating care with the patient as a partner. **Interpersonal & Communication Skills:** Creating and building rapport with patient.

16. Indirect Emotional Inquiry		

NURSE-

- Navigating and Responding to Emotion and Indirect Emotional Inquiry
- Emotion Handling and Empathy

Step	**Definition**	**Soft Skill Integration**	**Milestone/Core Competency**
Name	Stating the observed emotion to validate the patient's experience.	**12. Indirect Emotional Inquiry** ("It sounds like you're feeling overwhelmed.")	**ICS1:** Demonstrating empathy and responsiveness.
Understand	Legitimizing the emotion by acknowledging the patient's specific situation.	**10. Eliciting & Navigating Context** ("Given your work schedule, I can see why this is hard.")	**PC1:** Incorporating patient context into the history.
Respect	Praising the patient's efforts or strength in a difficult situation.	**2. Empathy & Compassion** ("You've done an incredible job caring for him.")	**PROF1:** Professional and ethical behavior.

Support	Explicitly stating your commitment to the patient's care and well-being.	**3. Relationship & Trust Building** ("We are going to figure this out together.")	**ICS1/ABIM:** Establishing a therapeutic alliance.
Explore	Asking follow-up questions to understand the depth of the emotional experience.	**7. Active Listening** and **9. Persistent Open-Ended Questioning**	**PC1:** Gathering accurate information through dialogue.

SHARE-

- Shared Decision Making, Collaborative phrasing, bridging relational gaps

Step	Definition	Soft Skill Integration	Milestone/Core Competency
Seek Participation	Inviting the patient to play an active role in the decision-making process.	**14. Collaborative Phrasing** ("I'd like for us to look at the options together.")	**ICS1:** Patient-centered communication and partnership.
Help Explore	Presenting various treatment options and discussing the	**11. Complex Clinical Reasoning** (Synthesizing data	**PC3/MK:** Clinical reasoning and Medical Knowledge.

	risks and benefits of each.	into clear trade-offs.)	
Assess Values	Helping the patient identify what matters most to them regarding their health and lifestyle.	**7. Active Listening** and **10. Eliciting & Navigating Context**	**SBP3:** System navigation for patient-centered care.
Reach Decision	Collaborating with the patient to finalize a plan that aligns with their preferences.	**15. Shared Decision Making** ("Which path feels most right for you?")	**ICS1:** Establishing a mutual management plan.
Evaluate	Reviewing the decision and discussing potential barriers to implementation.	**3. Relationship & Trust Building** (Checking for adherence and understanding.)	**PROF2:** Accountability for patient outcomes and follow-up.

Calgary Cambridge-

Phase	**Definition**	**Soft Skill Integration**	**Milestone/Core Competency**

Initiating	Establishing initial rapport and identifying the patient's reasons for the visit.	**5. Bedside Manner** and **13. Strategic Agenda Setting**	**ICS1:** Setting the foundation for a patient-centered encounter.
Gathering	Using specific techniques to elicit the patient’s narrative and medical history.	**8. Nonverbal Facilitation** and **9. Persistent Open-Ended Questioning**	**PC1:** Gathering accurate and comprehensive history.
Building	Developing the relationship through active rapport-building and sensitivity.	**4. Behavioral Adaptability** and **16. Bridging Relational Gaps**	**PROF1/ICS1:** Maintaining a therapeutic relationship throughout.
Explaining	Providing clear information and ensuring the patient understands the diagnosis.	**11. Complex Clinical Reasoning** and **14. Collaborative Phrasing**	**PC3:** Clinical reasoning; **ICS1:** Information sharing.
Closing	Summarizing the session and clarifying the safety net or follow-up plan.	**15. Shared Decision Making** and **3. Relationship & Trust Building**	**ICS1:** Clarity and commitment to the clinical plan.